

Homework 1

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Deadline: 2022-10-16 22:30:00, Beijing time

How to submit?

Email: YJ40202@163.com

Subject: [homework](#)

Attachments:

Eggs.java, Temperature.java, Divisible.java and Triangle.java

Note: Don't use package and don't submit compressed file.

How to submit?

Send

Save draft

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Mail template

Mass Mailing

Recipient : YJ40202<yj40202@163.com> email address

Subject : homework subject

Attach files Cloud attachment (100%, 2.72K uploaded, total file size2.72K)

Divisible.java

610B Uploaded

Delete

Eqqs.java

915B Uploaded

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Temperature.java

581B Uploaded

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Eggs.java

Meadowdale Dairy Farm sells organic brown eggs to local customers. It charges \$3.25 for a dozen eggs, or 45 cents for individual eggs that are not part of a dozen. Write a program that prompts a user for the number of eggs in the order and then display the amount owed with a full explanation. For example, typical output might be, You ordered 27 eggs. That's 2 dozen at \$3.25 per dozen and 3 loose eggs at 45 cents for a total of \$7.85.

Eggs.java

Sample Runs

```
Please enter the number of eggs you want to buy:10  
You ordered 10 eggs. That's 10 loose eggs at 45 cents for a total of $4.50
```

```
Please enter the number of eggs you want to buy:100  
You ordered 100 eggs. That's 8 dozen at $3.25 per dozen and 4 loose eggs  
at 45 cents for a total of $27.80
```

```
Please enter the number of eggs you want to buy:24  
You ordered 24 eggs. That's 2 dozen at $3.25 per dozen for a total of $6.50
```

Temperature.java

How cold is it outside? The temperature alone is not enough to provide the answer. Other factors including wind speed, relative humidity, and sunshine play important roles in determining coldness outside. In 2001, the National Weather Services (NWS) implemented the new wind-chill temperature to measure the coldness using the temperature and wind speed. The formula is

$$t_{wc} = 35.74 + 0.6215t_a - 35.75v^{0.16} + 0.4275t_av^{0.16}$$

where t_a is the outside temperature measured in degree Fahrenheit, v is the speed measured in miles per hour, and t_{wc} is the wind-chill temperature. The formula cannot be used for windspeed below 2 mph or temperature below -58 Fahrenheit or above 41 Fahrenheit.

Temperature.java

Write a program that prompts the user to enter a temperature between -58 Fahrenheit and 41 Fahrenheit and a wind speed greater than or equal to 2 then displays the wind-chill temperature. Use `Math.pow(a, b)` to compute $v^{0.16}$.

Sample Run

```
Please enter a temperature between -58 and 41 Fahrenheit:10
Please enter the wind speed (greater or equal to 2 mph):5
The wind-chill temperature is 1.23 Fahrenheit.
```

Divisible.java

Write a program that prompts the user to enter an integer and determine whether it is divisible by 5 and 6, whether it is divisible by 5 or 6, and whether it is divisible by 5 or 6, but not both.

Sample runs

```
Enter an integer:10
Is 10 divisible by 5 and 6? false
Is 10 divisible by 5 or 6? true
Is 10 divisible by 5 or 6? but not both? true
```

Triangle.java

Write a program that reads three edges (sides) for a triangle and classify it as Equilateral, Scalene, Isosceles if the input is valid. The input is valid if the sum of every pair of two edges is greater than the remaining edge. You may assume that the user inputs a positive value for the edges.

[Hint: Equilateral: All sides equal, Isosceles: Two sides equal, Scalene: All sides different]

Sample runs

```
Side1:3  
Side2:3  
Side3:9  
Output: Invalid input!
```

Triangle.java

```
Side1:3  
Side2:3  
Side3:3  
Output: Equilateral
```

```
Side1:10  
Side2:10  
Side3:1  
Output: Isosceles
```

```
Side1:3.2  
Side2:4.2  
Side3:5.1  
Output: Scalene
```